

Example 3

Analysis of CO₂ Emissions

Goals

1. Compute CO2 per 1000 people per year + region + source
 1. Goal: Create a longitudinal line graph
2. Compute Total CO2 per 1000 people per year + region
 1. Goal 1: Compute the top 10 overall combinations (year + region)
 2. Goal 2: Compute the top 3 regions per year for 2018+

Inspecting the raw data

Country Name	Country Code	Series Name	Series Code	2025 (YR202...	2024 (YR20...	2023 (YR20...	2022 (YR20...
Africa Eastern and Southern	AFE	Population, total	SP.POP.TOTL	788844284	769280888	7504913	731821393
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.MT.CE.AR5	..	634.4063	619.3657	618.5414
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.PC.CE.AR5	..	0.837679788408201	0.838104	0.858131540848484
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.AG.MT.CE.AR5	..	2.8442	2.6826	2.6826
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.BU.MT.CE.AR5	..	41.7037	41.177	41.177
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.FE.MT.CE.AR5	..	77.7873	74.5494	74.5494
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.IC.MT.CE.AR5	..	78.5775	73.9143	73.9143
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.IP.MT.CE.AR5	..	42.5011	39.1414	39.1414
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.PI.MT.CE.AR5	..	264.9669	250.9278	260.1308
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.TR.MT.CE.AR5	..	126.023	127.844	126.9431
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions	EN.GHG.CO2.WA.MT.CE.AR5	..	0.0034	0.0033	0.0032
Africa Western and Central	AFW	Population, total	SP.POP.TOTL	532809933	520655398	508318102	496366058
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.MT.CE.AR5	..	261.934	251.6472	259.3996
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.PC.CE.AR5	..	0.503085151918467	0.495058505707121	0.52259737711558
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.AG.MT.CE.AR5	..	1.514	1.4556	1.4155
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.BU.MT.CE.AR5	..	12.426	11.9931	12.3841
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.FE.MT.CE.AR5	..	34.9597	32.4378	32.9127
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.IC.MT.CE.AR5	..	21.7507	21.6229	22.5579
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.IP.MT.CE.AR5	..	33.7165	33.2706	34.4196
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.PI.MT.CE.AR5	..	45.5711	42.9397	44.1713
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.TR.MT.CE.AR5	..	111.9837	107.9158	111.5267
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions	EN.GHG.CO2.WA.MT.CE.AR5	..	0.0124	0.0119	0.0115
Arab World	ARB	Population, total	SP.POP.TOTL	503356133	492612632	482105978	471352066
Arab World	ARB	Carbon dioxide (CO2) emissions	EN.GHG.CO2.MT.CE.AR5	..	2229.854	2171.3007	2106.1097
Arab World	ARB	Carbon dioxide (CO2) emissions	EN.GHG.CO2.PC.CE.AR5	..	4.57571631886073	4.55256379383056	4.51655911861293
Arab World	ARB	Carbon dioxide (CO2) emissions	EN.GHG.CO2.AG.MT.CE.AR5	..	3.9359	3.9489	4.017

Data in NAME
→ Stack

Drop

Drop

Variable in COLUMN
→ Unstack

Remove the CODE columns

World_Bank_C... CO2 raw
Drop Code Columns

Click ...

Clean 9 70 fields 3K rows

Identify Duplicate Rows Rename Fields... Pivot Columns to Rows Merge Fields ... 2 Recommendations

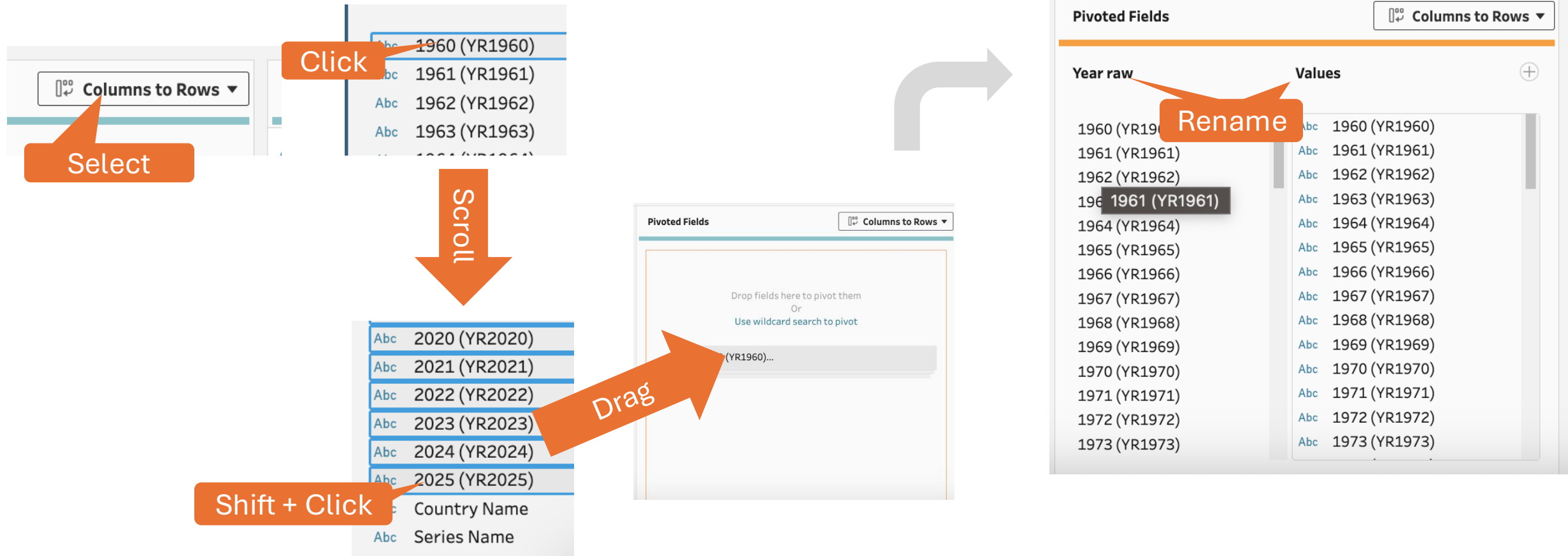
Search

Country Name	Country Code	Series Name	Series Code	(YR20...	2024 (YR20...	2023 (YR20...
Africa Eastern and Southern	AFE	Population, total	SP.POP.TOTL	788844284	769280888	750491370
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions (total) excluding LULUCF (Mt CO2e)	EN.GHG.CO2.MT.CE.AR5	..	634.4063	619.3657
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions excluding LULUCF per capita (kg CO2e)	EN.GHG.CO2.PC.CE.AR5	..	0.837679788408201	0.838104192853686
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions from Agriculture (Mt CO2e)	EN.GHG.CO2.AG.MT.CE.AR5	..	2.8442	2.7646
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions from Building (Energy) (Mt CO2e)	EN.GHG.CO2.BU.MT.CE.AR5	..	41.7037	41.7476
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions from Fugitive Emissions (Energy) (Mt CO2e)	EN.GHG.CO2.FE.MT.CE.AR5	..	77.7873	77.6554
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions from Industrial Combustion (Energy) (Mt CO2e)	EN.GHG.CO2.IC.MT.CE.AR5	..	78.5775	77.7505
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions from Industrial Processes (Energy) (Mt CO2e)	EN.GHG.CO2.IP.MT.CE.AR5	..	42.5011	40.6725
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions from Power Industry (Energy) (Mt CO2e)	EN.GHG.CO2.PI.MT.CE.AR5	..	264.9669	250.9278
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions from Transport (Energy) (Mt CO2e)	EN.GHG.CO2.TR.MT.CE.AR5	..	126.023	127.844
Africa Eastern and Southern	AFE	Carbon dioxide (CO2) emissions from Waste (Mt CO2e)	EN.GHG.CO2.WA.MT.CE.AR5	..	0.0034	0.0033
Africa Western and Central	AFW	Population, total	SP.POP.TOTL	532809933	520655398	508318102
Africa Western and Central	AFW	Carbon dioxide (CO2) emissions (total) excluding LULUCF (Mt CO2e)	EN.GHG.CO2.MT.CE.AR5	..	261.934	251.6472

CTRL/CMD +
Click
to select

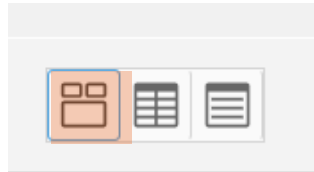
Stack the YEAR columns

Pivot → Columns to Rows



Goal: Clean up the VARIABLE names

Solution: Click and edit the labels



Series Name 12

Clean these up!

<i>null</i>
Carbon dioxide (CO2) emissions (total) excluding LULUCF (Mt CO2e)
Carbon dioxide (CO2) emissions excluding LULUCF per capita (t CO2e/capita)
Carbon dioxide (CO2) emissions from Agriculture (Mt CO2e)
Carbon dioxide (CO2) emissions from Building (Energy) (Mt CO2e)
Carbon dioxide (CO2) emissions from Fugitive Emissions (Energy) (Mt CO2e)
Carbon dioxide (CO2) emissions from Industrial Combustion (Energy) (Mt CO2e)
Carbon dioxide (CO2) emissions from Industrial Processes (Mt CO2e)
Carbon dioxide (CO2) emissions from Power Industry (Energy) (Mt CO2e)
Carbon dioxide (CO2) emissions from Transport (Energy) (Mt CO2e)
Carbon dioxide (CO2) emissions from Waste (Mt CO2e)
Population, total

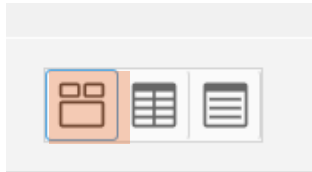
Mantra. Keep the simple things simple!

Strategies.

1. **Click and Edit for small changes**
2. Use the MENU options when convenient.
3. Only resort to code when necessary

Goal: Clean up the VARIABLE names

Solution: Click and edit the labels



Series Name 12 ≡ ▾ 🔍 ...

null

- Carbon dioxide (CO2) emissions (total) excluding LULUCF (Mt CO2e)
- Carbon dioxide (CO2) emissions excluding LULUCF per capita (t CO2e/capita)
- Carbon dioxide (CO2) emissions from Agriculture (Mt CO2e)
- Carbon dioxide (CO2) emissions from Building (Energy) (Mt CO2e)
- Carbon dioxide (CO2) emissions from Fugitive Emissions (Energy) (Mt CO2e)
- Carbon dioxide (CO2) emissions from Industrial Combustion (Energy) (Mt CO2e)
- Carbon dioxide (CO2) emissions from Industrial Processes (Mt CO2e)
- Carbon dioxide (CO2) emissions from Power Industry (Energy) (Mt CO2e)
- Carbon dioxide (CO2) emissions from Transport (Energy) (Mt CO2e)
- Carbon dioxide (CO2) emissions from Waste (Mt CO2e)
- Population, total

Rename

Variables 10 🔗

- Agriculture
- Building
- Fugitive Emissions
- Industrial Combustion
- Industrial Processes
- Population, total
- Power Industry
- Total CO2
- 🔗 Transport
- 🔗 Waste

Click + Edit

Goal: Clean up the YEAR RAW column

Solution: Split the string



Year raw
1960(YR1960)
1961(YR1961)
1962(YR1962)
1963(YR1963)
1964(YR1964)
1965(YR1965)
1966(YR1966)
1967(YR1967)
1968(YR1968)
1969(YR1969)
1970(YR1970)
1971(YR1971)
1972(YR1972)

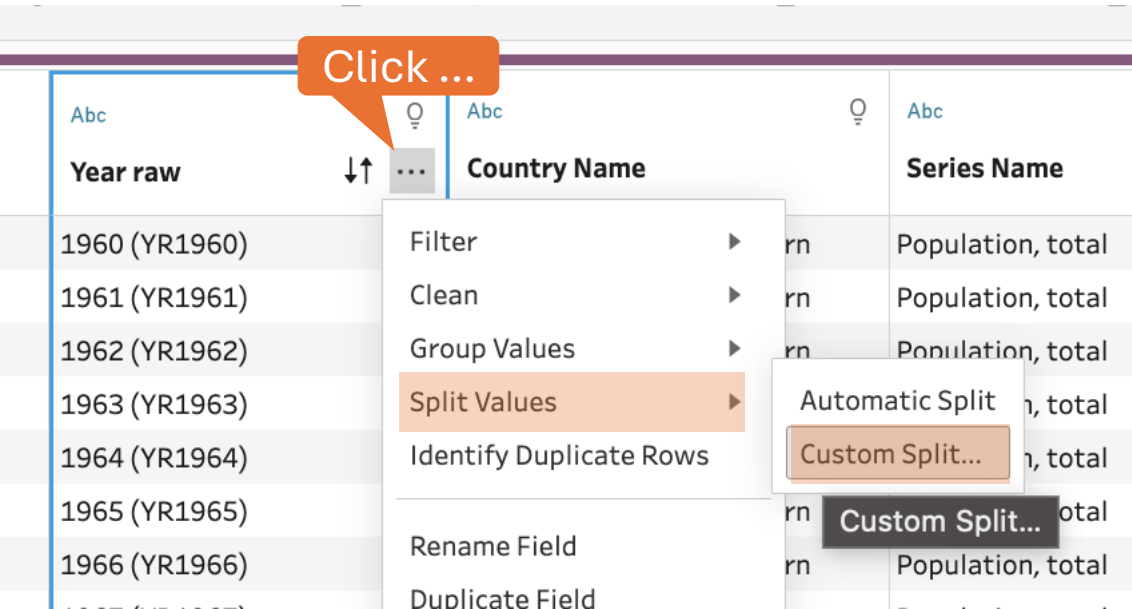
Mantra. Keep the simple things simple!

Strategies.

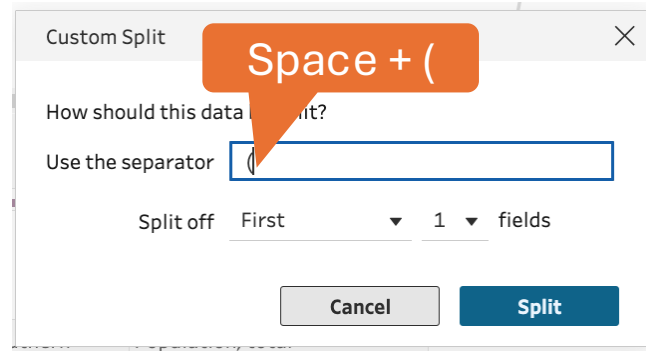
1. Click and Edit for small changes
2. Use the MENU options when convenient.
3. Only resort to code when necessary

Cleaning up the YEAR column

Split → Rename → Cast as whole number → Drop YEAR RAW



Year raw	Country Name	Series Name
1960 (YR1960)		Population, total
1961 (YR1961)		Population, total
1962 (YR1962)		Population, total
1963 (YR1963)		Population, total
1964 (YR1964)		Population, total
1965 (YR1965)		Population, total
1966 (YR1966)		Population, total

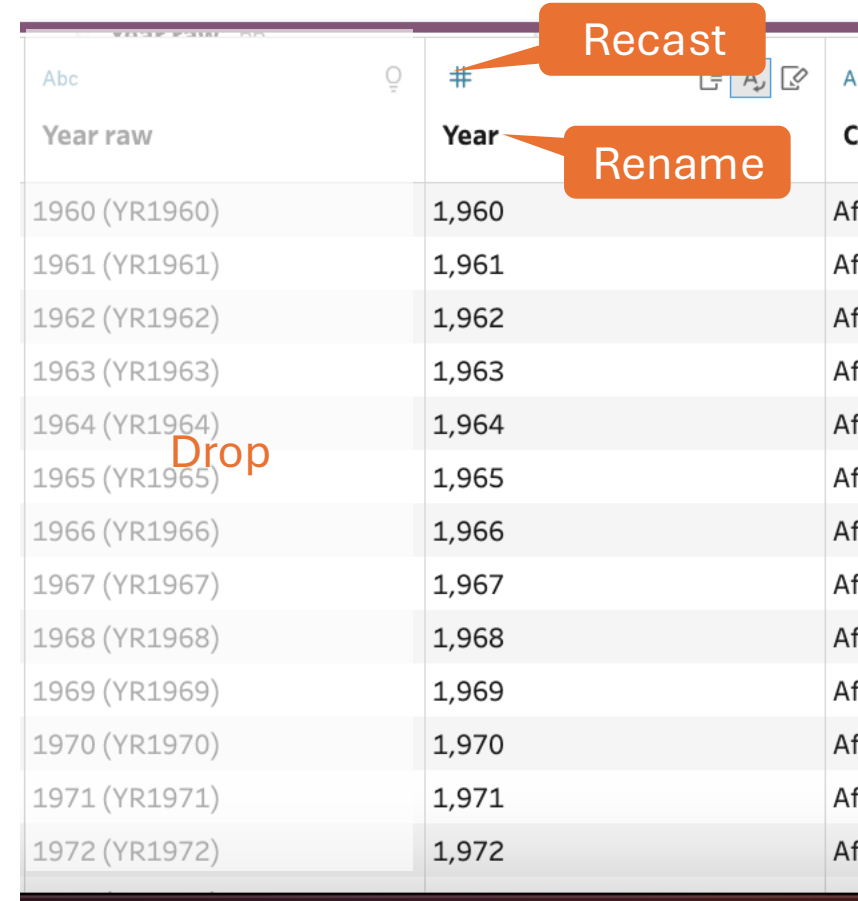


Custom Split

How should this data be split?

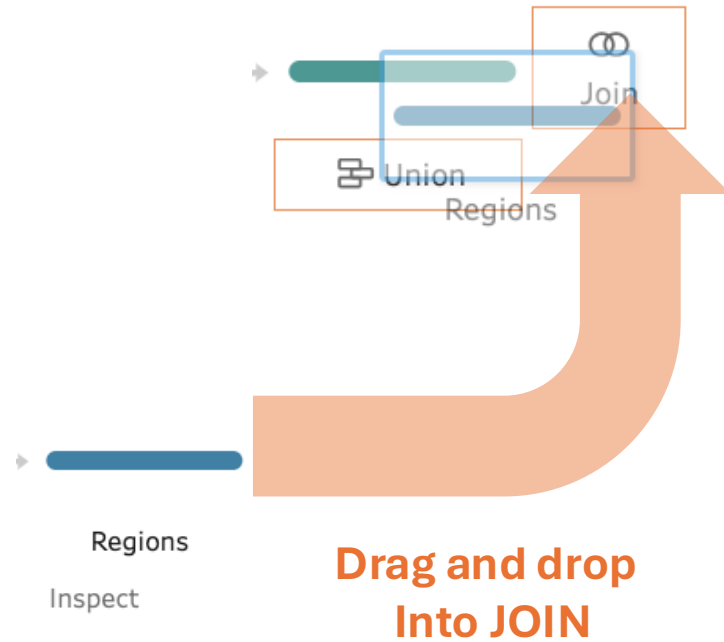
Use the separator

Split off fields



Year raw	Year	Drop
1960 (YR1960)	1,960	1960 (YR1960)
1961 (YR1961)	1,961	1961 (YR1961)
1962 (YR1962)	1,962	1962 (YR1962)
1963 (YR1963)	1,963	1963 (YR1963)
1964 (YR1964)	1,964	1964 (YR1964)
1965 (YR1965)	1,965	1965 (YR1965)
1966 (YR1966)	1,966	1966 (YR1966)
1967 (YR1967)	1,967	1967 (YR1967)
1968 (YR1968)	1,968	1968 (YR1968)
1969 (YR1969)	1,969	1969 (YR1969)
1970 (YR1970)	1,970	1970 (YR1970)
1971 (YR1971)	1,971	1971 (YR1971)
1972 (YR1972)	1,972	1972 (YR1972)

JOIN on the region names



Join 2 6 fields 867K rows

Filter Values... Identify Duplicate Rows Create Calculated Field...

Applied Join Clauses

Clean 9 Regions

Country Name = Country

Join Type : Inner

Click the graphic to change the join type.

Clean 9 Regions

INNER join

Summary of Join Results

Click the bar segments to view

Mismatched values

	Included	Excluded
Clean 9	144,474	48,246

Join Clauses

Show only mismatched values

Clean 9

↑ Country Name

null

Afghanistan

Africa Eastern and Southern

Africa Western and Central

Albania

Algeria

American S...

Regions

↑ Country

Cyprus

Czech Republic

Denmark

Djibouti

Dominica

Dominican Republic

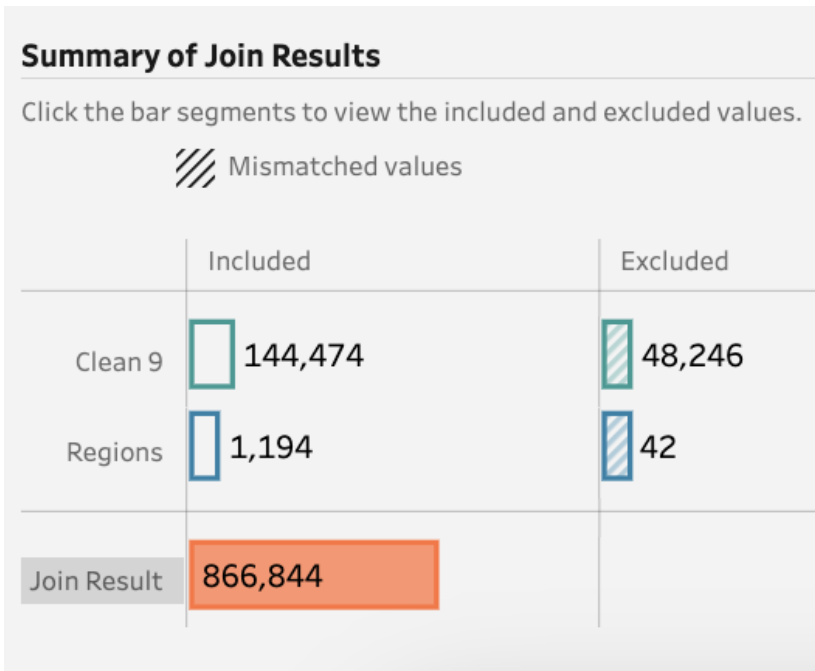
Ecuador

Fix mismatch
In this column

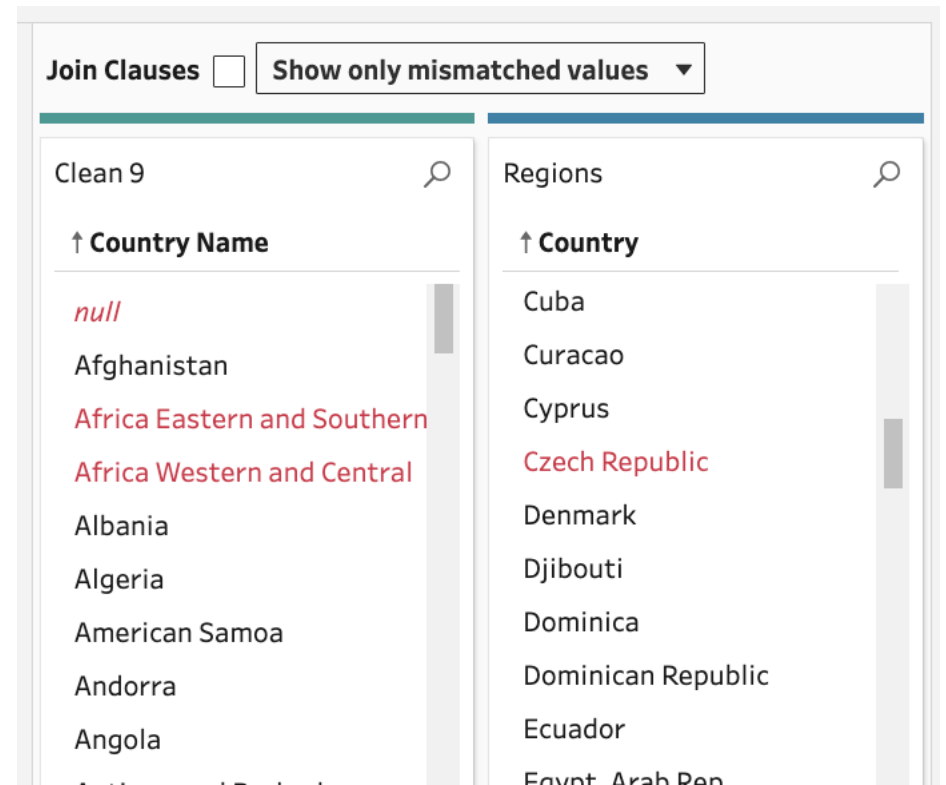
INNER join
→ Remove regions totals
(GOOD)

Tableau Prep → Joins done right!

Great summary of results



Find & Fix Mismatches



Goals

1. Compute CO2 per 1000 people per year + region + source

1. Goal: Create a longitudinal line graph

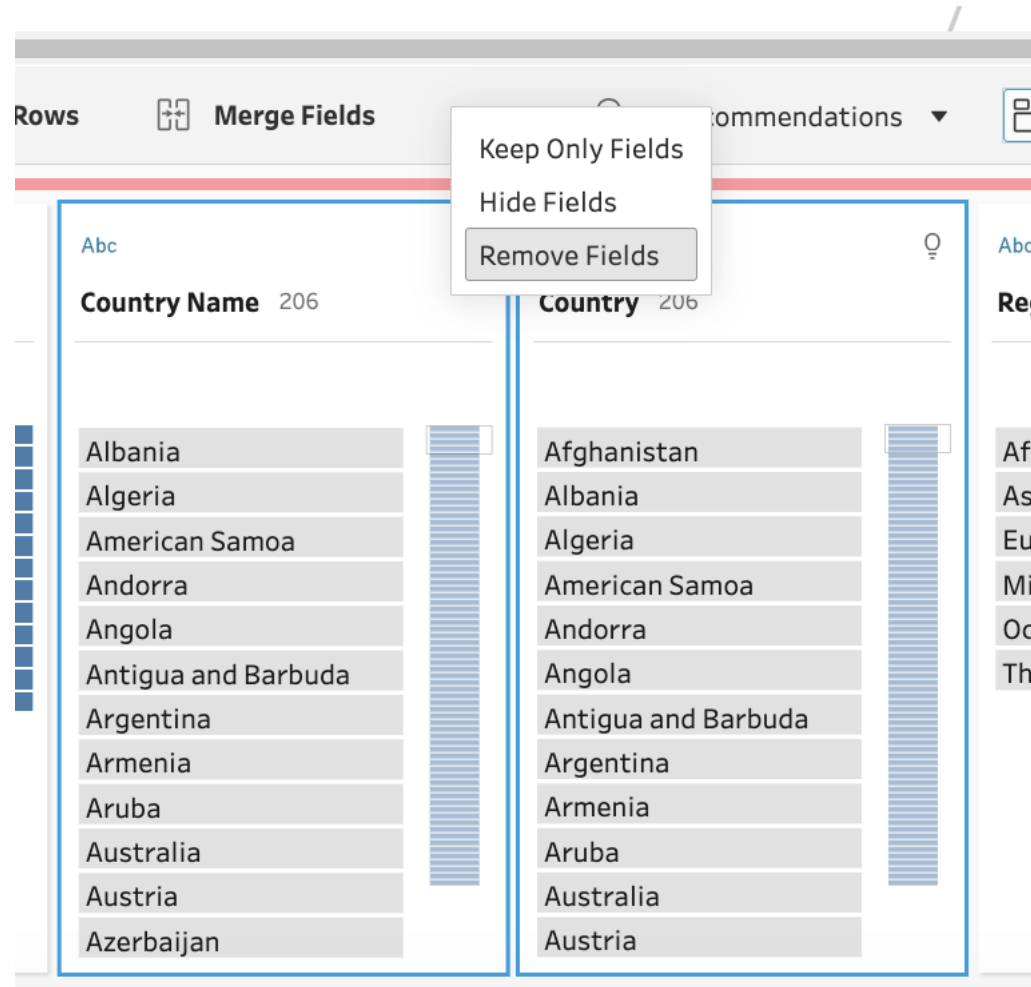
2. Compute Total CO2 per 1000 people per year + region

1. Goal 1: Compute the top 10 overall combinations (year + region)

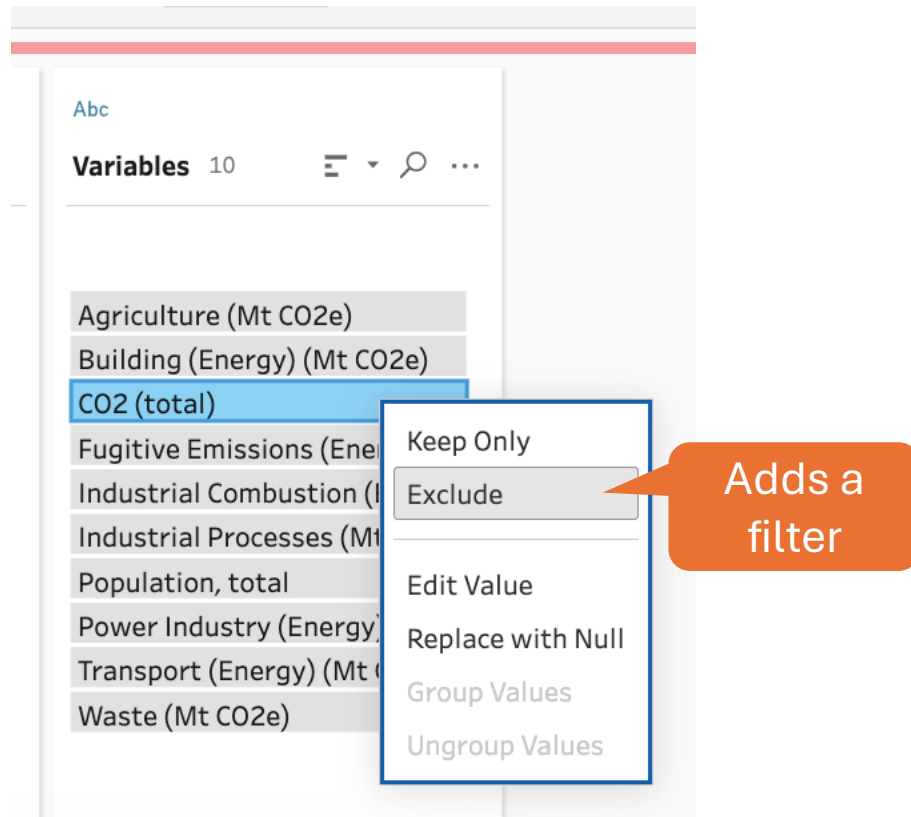
2. Goal 2: Compute the top 3 regions per year for 2018-2020

We group by region

→ Drop the country names



Filter out the total emissions



Mantra. Keep the simple things simple!

Strategies.

1. Click and Edit for small changes
2. Use the MENU options when convenient.
3. Only resort to code when necessary

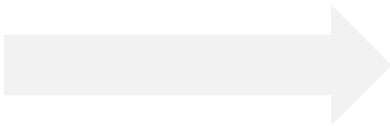
We have a problem

Problem. We need to pull the population totals into a column to compute CO2 per 1000 people

Abc

Variables 9

Agriculture (Mt CO2e)
Building (Energy) (Mt CO2e)
Fugitive Emissions (Energy) (Mt CO2e)
Industrial Combustion (Energy) (Mt CO2e)
Industrial Processes (Mt CO2e)
Population, total
Power Industry (Energy) (Mt CO2e)
Transport (Energy) (Mt CO2e)
Waste (Mt CO2e)



CO2 by type
Unstack variables

CO2 by type
Stack CO2 sources

Abc	#
Type	Population, ...
Agriculture (Mt CO2e)	21,801,040,170
Building (Energy) (Mt CO2e)	21,801,040,170
Waste (Mt CO2e)	21,801,040,170
Industrial Combustion (Energy) (Mt CO2e)	21,801,040,170
Fugitive Emissions (Energy) (Mt CO2e)	21,801,040,170
Industrial Processes (Mt CO2e)	21,801,040,170
Power Industry (Energy) (Mt CO2e)	21,801,040,170
Transport (Energy) (Mt CO2e)	21,801,040,170
Industrial Processes (Mt CO2e)	1,146,638,208
Fugitive Emissions (Energy) (Mt CO2e)	1,146,638,208
Building (Energy) (Mt CO2e)	1,146,638,208

Solution. Unstack Variable → Stack CO2 sources

Computing a per-region rate

Compute totals → Divide



Pivoted Fields Rows to Columns ▾

Variable ✕

- Agriculture
- Building
- Fugitive Emissions
- Industrial Combustion
- Industrial Processes
- Population, total
- Power Industry
- Transport
- Waste

Field to aggregate for new columns ✕

#	SUM	Values
---	-----	--------

1) Compute Totals
with pivot

2) Rate after
aggregation (pivot)

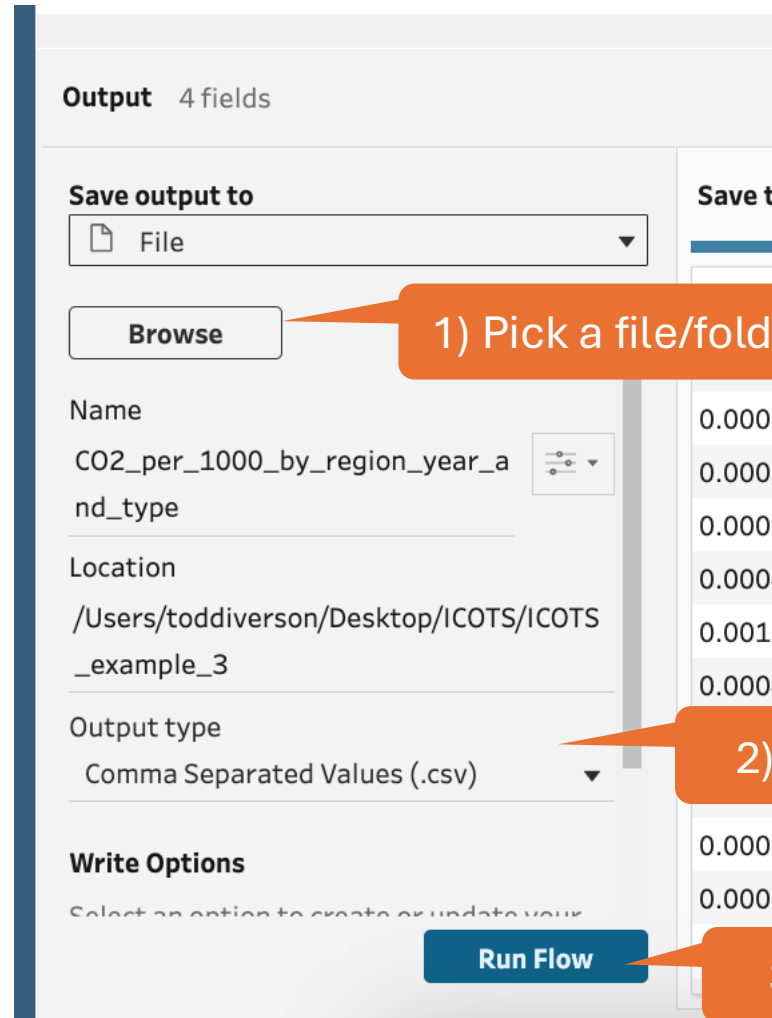
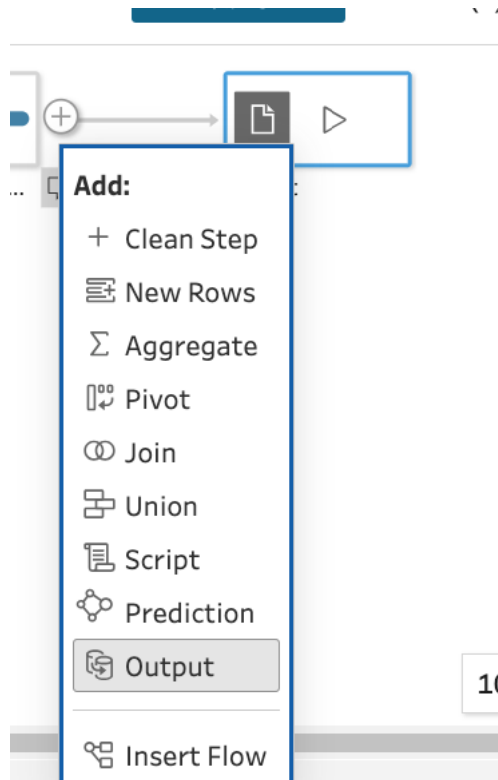
Edit Field

Field Name

C02 per 1000 people

`[C02 emissions]/([Population, total]/1000)`

Outputting a CSV file



1) Pick a file/folder

2) Switch to CSV

3) Click RUN to write the output file

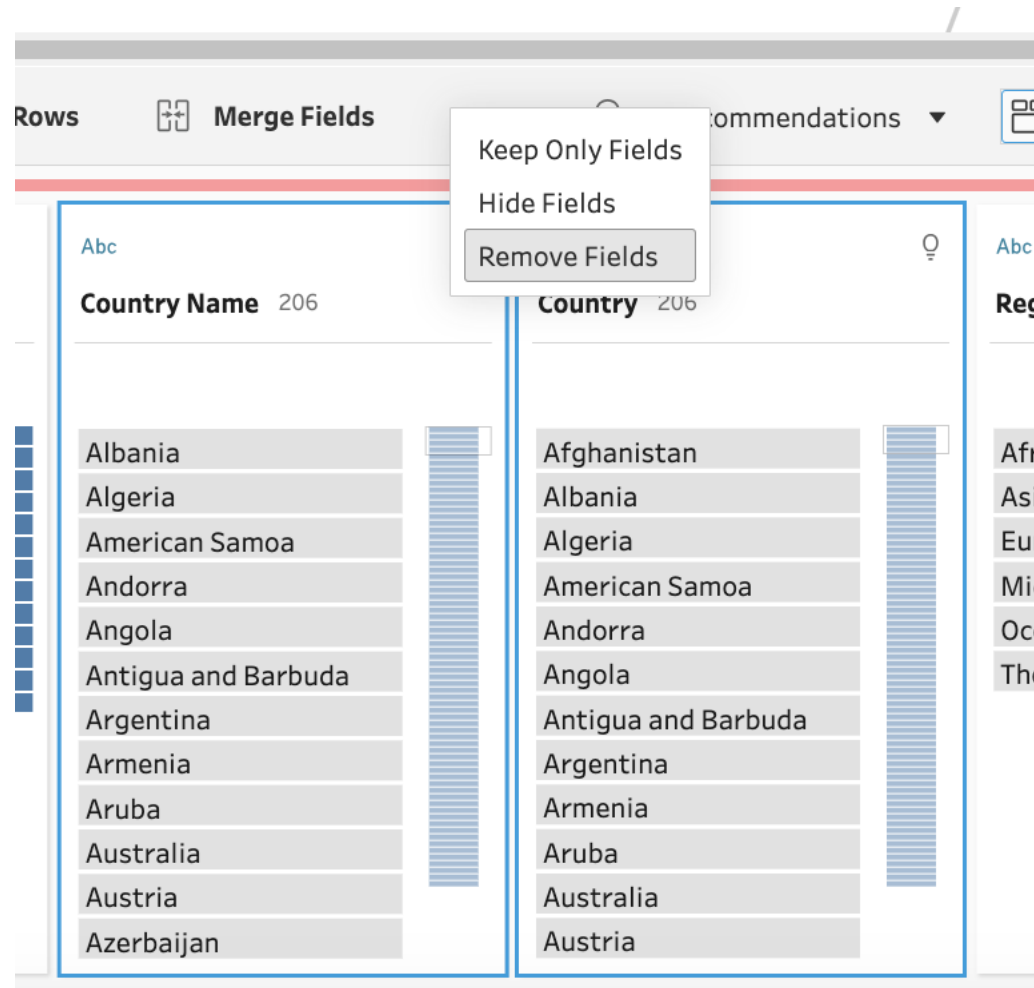
Goals

1. Compute CO2 per 1000 people per year + region + source
 1. Goal: Create a longitudinal line graph
2. **Compute Total CO2 per 1000 people per year + region**
 1. **Goal 1: Compute the top 10 overall combinations (year + region)**
 2. **Goal 2: Compute the top 3 regions per year for 2018+**

Drop the country names

Goal 1: Compute the top 10 overall combinations (year + region)

Goal 2: Compute the top 3 regions per year for 2018+



Mantra. Keep the simple things simple!

Strategies.

1. Click and Edit for small changes
2. Use the **MENU options when convenient.**
3. Only resort to code when necessary

Filter out the total

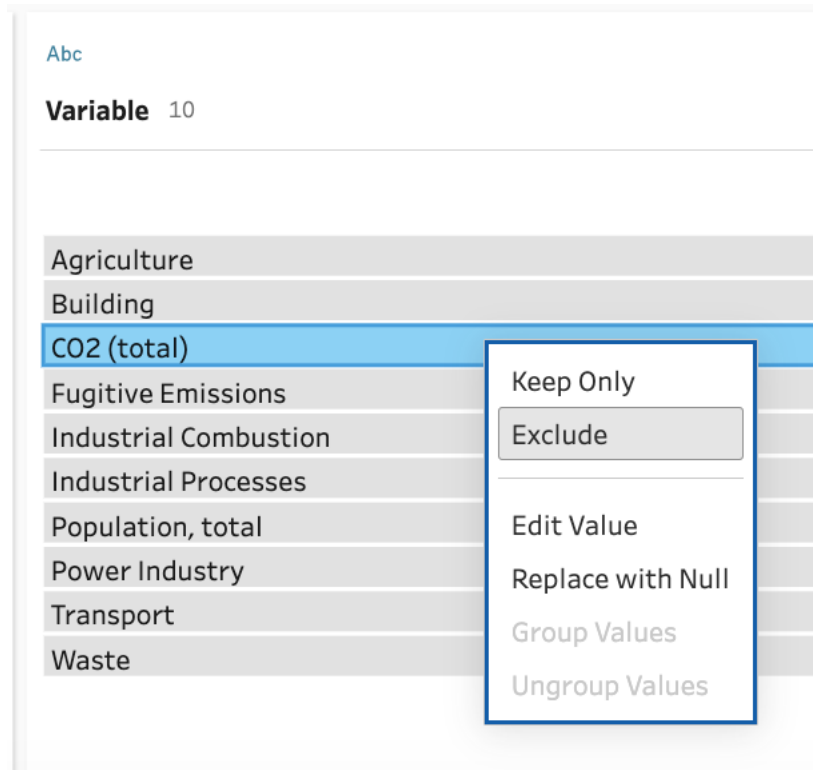
Goal 1: Compute the top 10 overall combinations (year + region)

Goal 2: Compute the top 3 regions per year for 2018+

Mantra. Keep the simple things simple!

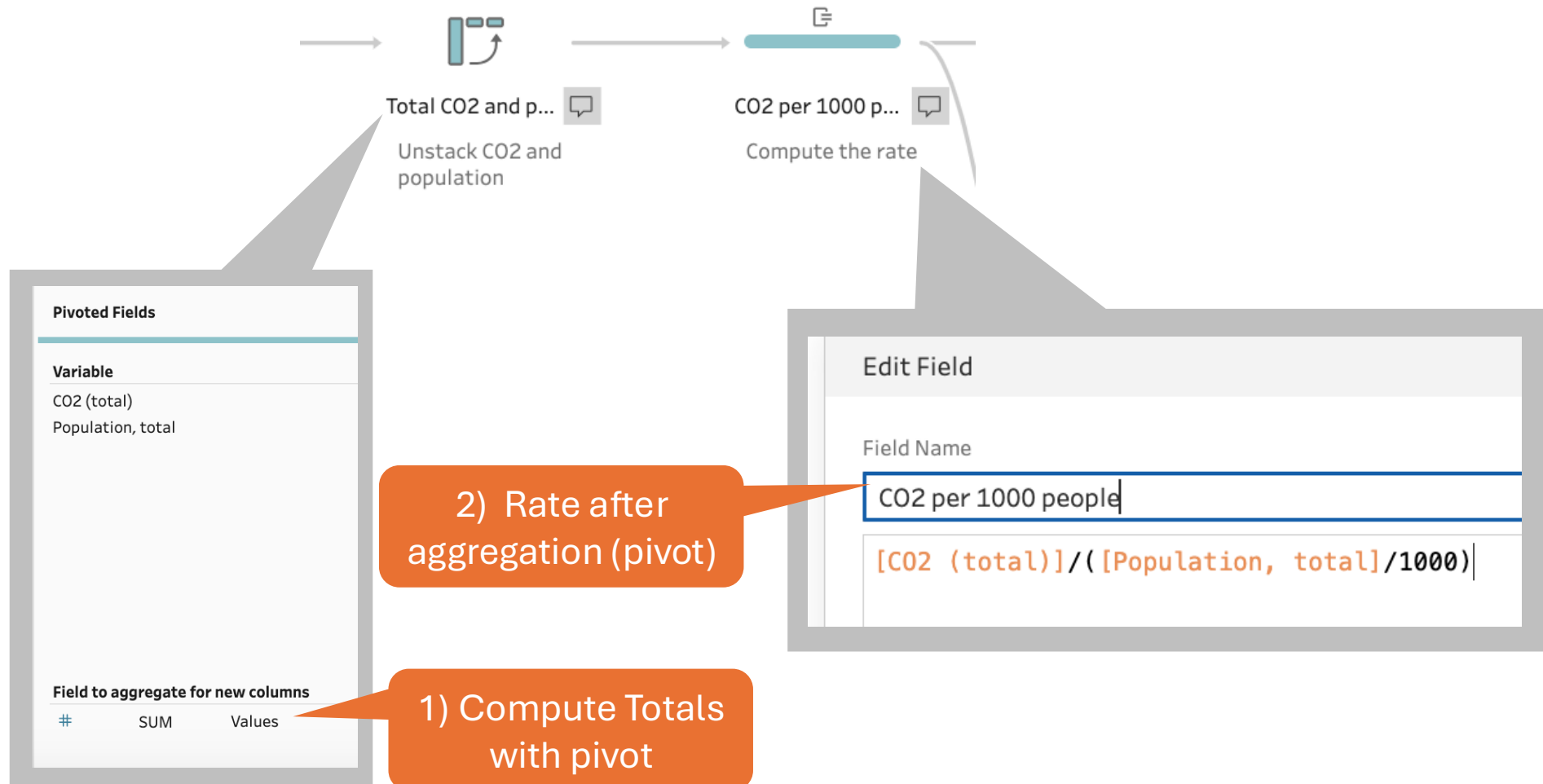
Strategies.

1. Click and Edit for small changes
2. Use the **MENU options when convenient.**
3. Only resort to code when necessary



Computing a per-region rate

Compute totals → Divide



Split → Rank → Filter

Goal 1: Compute the top 10 overall combinations (year + region)
Goal 2: Compute the top 3 regions per year for 2018+

